

Intelligence

Lecture 2

Origins of IQ Testing

Dr Steven Poulter

Key Themes

- Is there one intelligence or many?
- Origins of IQ Testing
- Distribution of IQ scores
- Importance of age in intelligence testing



Glossary of Key Terms

Intelligence

A broad mental ability that helps us learn, reason, solve problems and adapt to new situations. Hard to define exactly; often treated as “general mental ability”.

Mental abilities

Specific things the mind can do, such as remembering, reasoning, using language, solving problems, or manipulating shapes in space.

Operationalisation / Operational definition

Turning a fuzzy idea (like “intelligence” or “aggression”) into something we can actually measure (e.g. score on a test, number of shoves, answers to certain questions).

General intelligence (g, g-factor)

A statistical factor that captures what is common across lots of different mental tests. If you tend to do well on many different types of tests, you are said to have higher g.

Specific intelligences

Abilities that are more narrow or specialised (e.g. verbal, spatial, mathematical, mechanical). You can be stronger in one area than another.

Savant syndrome

A condition where someone demonstrates exceptional aptitude in one domain, such as art or mathematics, with such aptitude often coinciding with some form of social or intellectual impairment.

IQ (Intelligence Quotient)

A single score meant to summarise someone's performance on an 'intelligence test' compared to the general population. Modern IQ scores are based on how far above or below the average (100) a person scores.

Mental age

In early IQ tests: the age level at which a child is performing. For example, if a 7-year-old passes most items typical of 9-year-olds, we might say they have a mental age of 9.

Chronological / physical age

Your actual age in years (e.g. 10 years old), used to compare with mental age in early IQ formulas.

IQ ratio (Stern's formula)

An early way of calculating IQ in children:

$$\text{IQ} = \text{Mental age} / \text{Physical age} \times 100$$

e.g. mental age 8 / physical age 10 $\times$ 100 = IQ 80.

Norm-referenced / relative measure

Scores are meaningful only compared to a large group (the "norm" sample) of people of the same age (or other criterion). IQ is defined relative to others, not as an absolute quantity.

Ordinal measure

A scale that tells you the order (who scored higher or lower), but not “how much” more ability someone has. IQ is treated as ordinal: 120 is higher than 100, but not “20% more intelligence”.

Independent criterion (of intelligence)

An external measure that we treat as a standard to judge a test against—for example, school grades or job performance. Good intelligence tests should correlate with such criteria.

Consequential behaviour

Real-world outcomes we care about, such as school success or job performance, that we might want intelligence tests to predict.

Standardised test

A test that is given and scored in the same way for everyone, under the same conditions, so scores can be compared fairly across people.

Normal distribution / “bell curve”

A common pattern of scores where most people are in the middle, and fewer people are very high or very low. IQ scores in a large sample tend to form a bell-shaped curve.

Mean

The average score. For IQ, by definition, the mean is set to 100.

Standard deviation (SD)

A measure of how spread out the scores are around the mean. For IQ, SD = 15. Being 1 SD above the mean is IQ 115; 2 SD above is IQ 130, etc.

Percentile

Shows the percentage of people who score below you. Being in the top 2.5% means you scored higher than about 97.5% of people your age.

Average IQ

Usually defined as IQ 85–115 (about 68% of people fall in this range).

Gifted (in IQ terms)

Defined here as IQ above about 145; roughly less than the top 1% of the population, depending on the cut-off used.

Intellectual disability (in IQ terms)

Traditionally defined as IQ below about 70 (~2.5%), usually combined with difficulties in everyday functioning. Note IQ score alone not enough to define this term in clinical/educational settings.

Historical figures & tests

Sir Francis Galton (1822–1911)

Early pioneer of measuring individual differences; founded differential psychology and helped start psychometrics. Tried to measure intelligence via basic sensory and reaction time tasks. Coined “nature vs nurture”.

Differential psychology

The branch of psychology that studies how and why individuals differ from one another in traits such as intelligence, personality, etc.

Psychometrics

The science of psychological measurement—designing, testing and analysing questionnaires and tests (e.g. IQ tests) to ensure they are reliable and valid.

Anthropometric Laboratory

Galton’s London lab where he collected large amounts of data (around 9,000 people) on physical and sensory measures, hoping to relate them to mental ability.

James McKeen Cattell (1860–1944)

Psychologist who, following Galton, developed the first set of “mental tests” (e.g. reaction time, weight discrimination) to measure individual differences.

Clark Wissler (1870–1947)

Used correlation statistics to show Cattell's mental tests did not predict college grades (and didn't even correlate with each other), which undermined that early approach to IQ.

Alfred Binet (1857–1911)

Commissioned by the French government to develop the first practical intelligence test for children (the Binet–Simon scale) to identify those needing extra support. Focused on everyday tasks and age-appropriate items.

Binet–Simon scale

Early intelligence test with sets of tasks grouped by age (e.g. 4-year-old tasks, 6-year-old tasks). The pattern of items a child could pass was used to estimate their mental age.

Lewis Terman (1877–1956)

American psychologist who revised Binet's test into the **Stanford–Binet** test, expanded it, and helped popularise IQ testing in the US.

Stanford–Binet test

A major revision of the Binet–Simon scale produced at Stanford University; became one of the standard IQ tests and is still used today (in updated forms).

William Stern (1871–1938)

Psychologist who introduced the term “Intelligence Quotient” (IQ) and proposed the mental-age/chronological-age ratio formula for children's IQ.

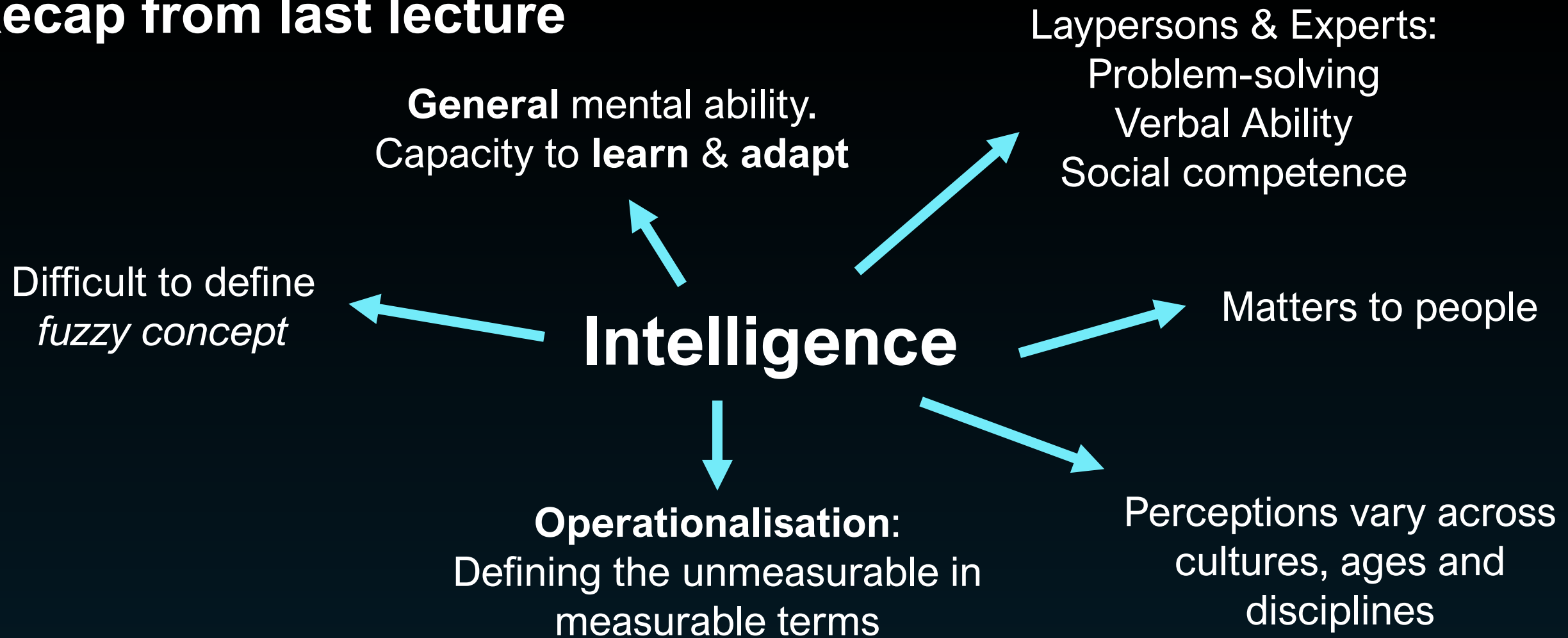
Robert Yerkes (1876–1956)

Psychologist who, with colleagues, developed large-scale IQ tests (Army Alpha and Beta) for the US Army during World War I, boosting the popularity of mental testing.

Army Alpha and Beta tests

Group intelligence tests used by the US Army: Alpha for literate recruits, Beta for illiterate or non-English speakers. Multiple-choice, timed tasks given to huge numbers of people.

Recap from last lecture



The important point is not whether what we measure can appropriately be labelled 'intelligence', but whether we have discovered something worth measuring.

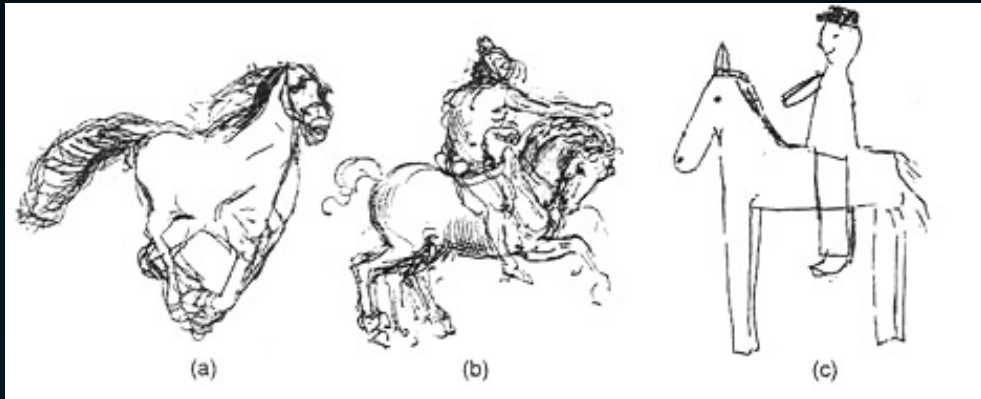
Miles, (1957, p. 159)

What is intelligence?

Savant syndrome – special ability usually accompanied by prodigious memory

Approx. 1 in 10 persons with autistic disorder show some savant skills *Treffert (2009)*

Nadia Chomyn (*Selfe 1978*)



Drawing ability at 5-years-old

Brittany Maier



Musical ability

Kim Peek



Memory ability

‘Savant’ or ‘knowledgeable person’ derived from French word **savoir** meaning ‘to know’.

What is intelligence?

Are these specific savant abilities evidence of high intelligence?

“No, in that someone with savant syndrome typically has a low IQ score, developmental disabilities and often cannot care for themselves”

Haier (2022)

“A savant is a person (not necessarily a child) who displays an island of exceptional mental performance in a sea of disability”

Feldman & Morelock (2012)

Extraordinary but narrow mental ability is not what is usually considered evidence of high intelligence by many researchers

What is intelligence?

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No, in that someone with savant syndrome typically has a low IQ score, developmental disabilities and often cannot care for themselves
Haier (2022)

“A savant is a person (not necessarily a child) who displays an island of exceptional mental performance in a sea of disability”
Feldman & Morelock 2012

However, savant research raises questions on the existence of multiple intelligences in the areas where savants appear, e.g. music, mathematics, art, mnemonics etc., even in the face of low IQ scores

What is intelligence?



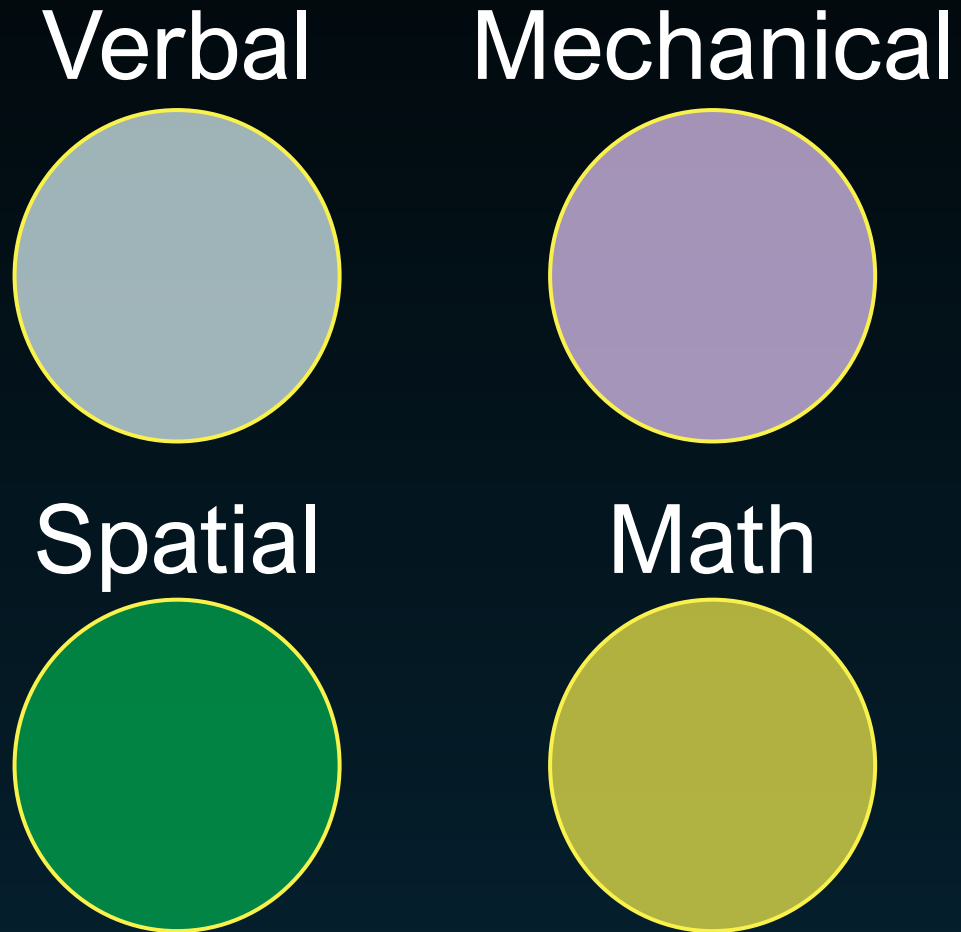
Is **athletic ability** a kind of intelligence?



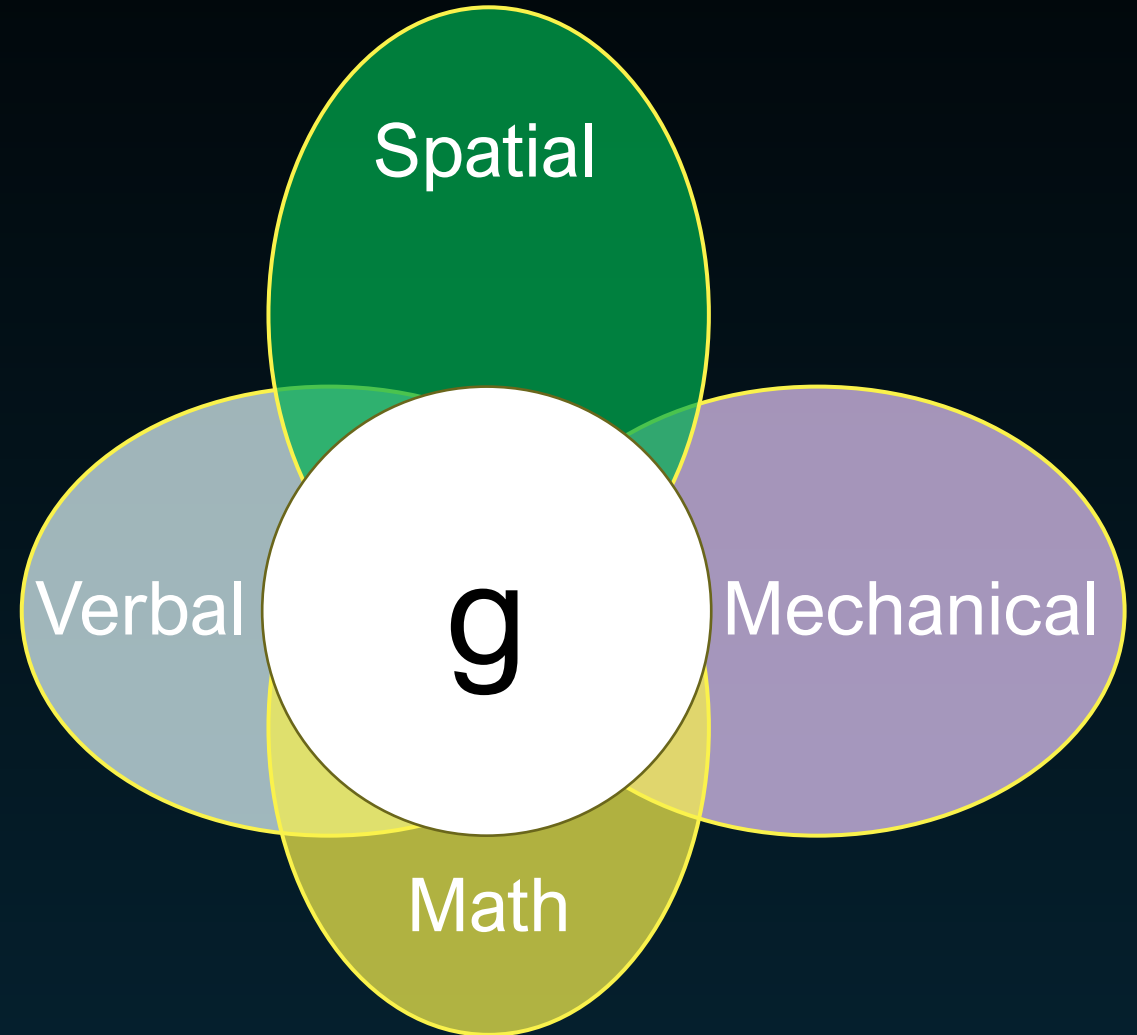
Watson IBM Computer
similar kind of intelligence?

Is there one intelligence or many?

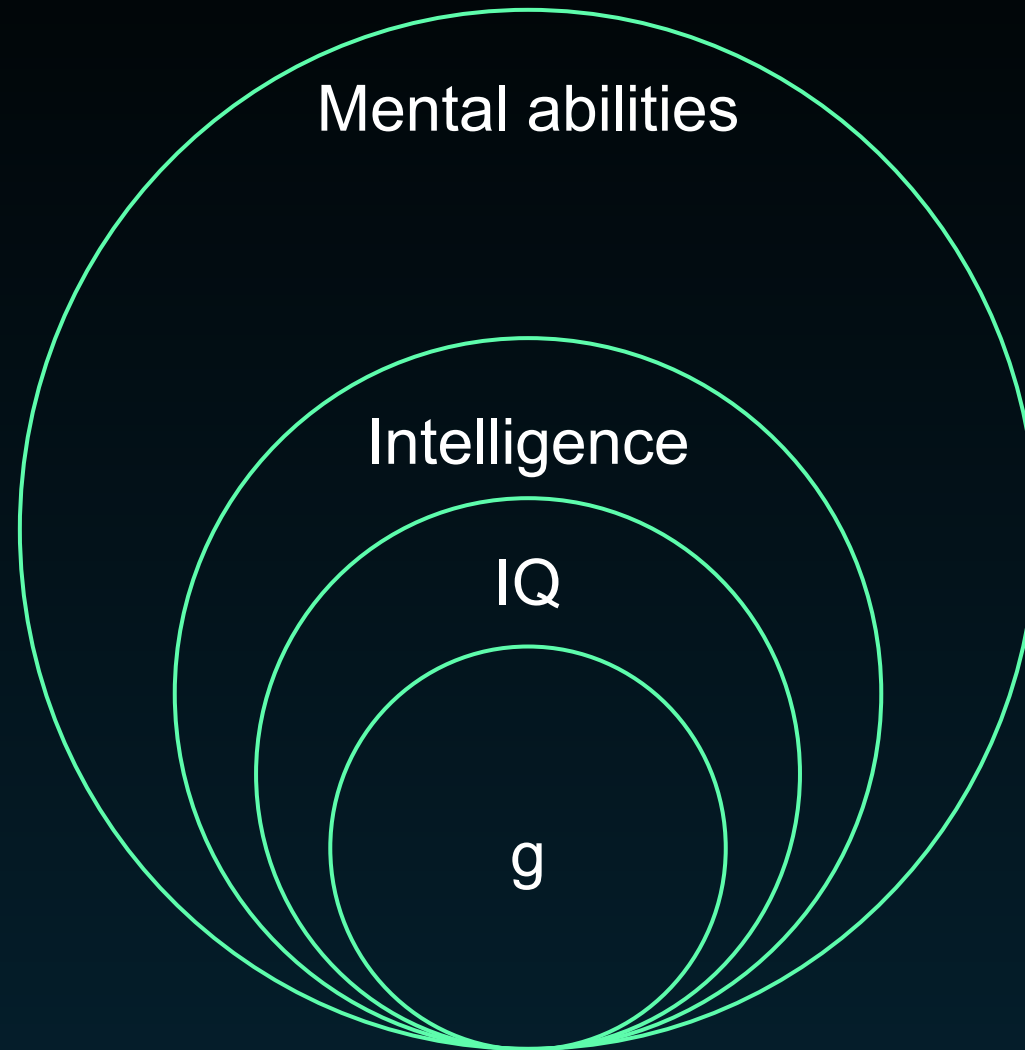
Specific Intelligences



General Intelligence (g)



Defining terms



‘Intelligence’ – ambiguous for scientific use, so....

2 common substitutes are used:

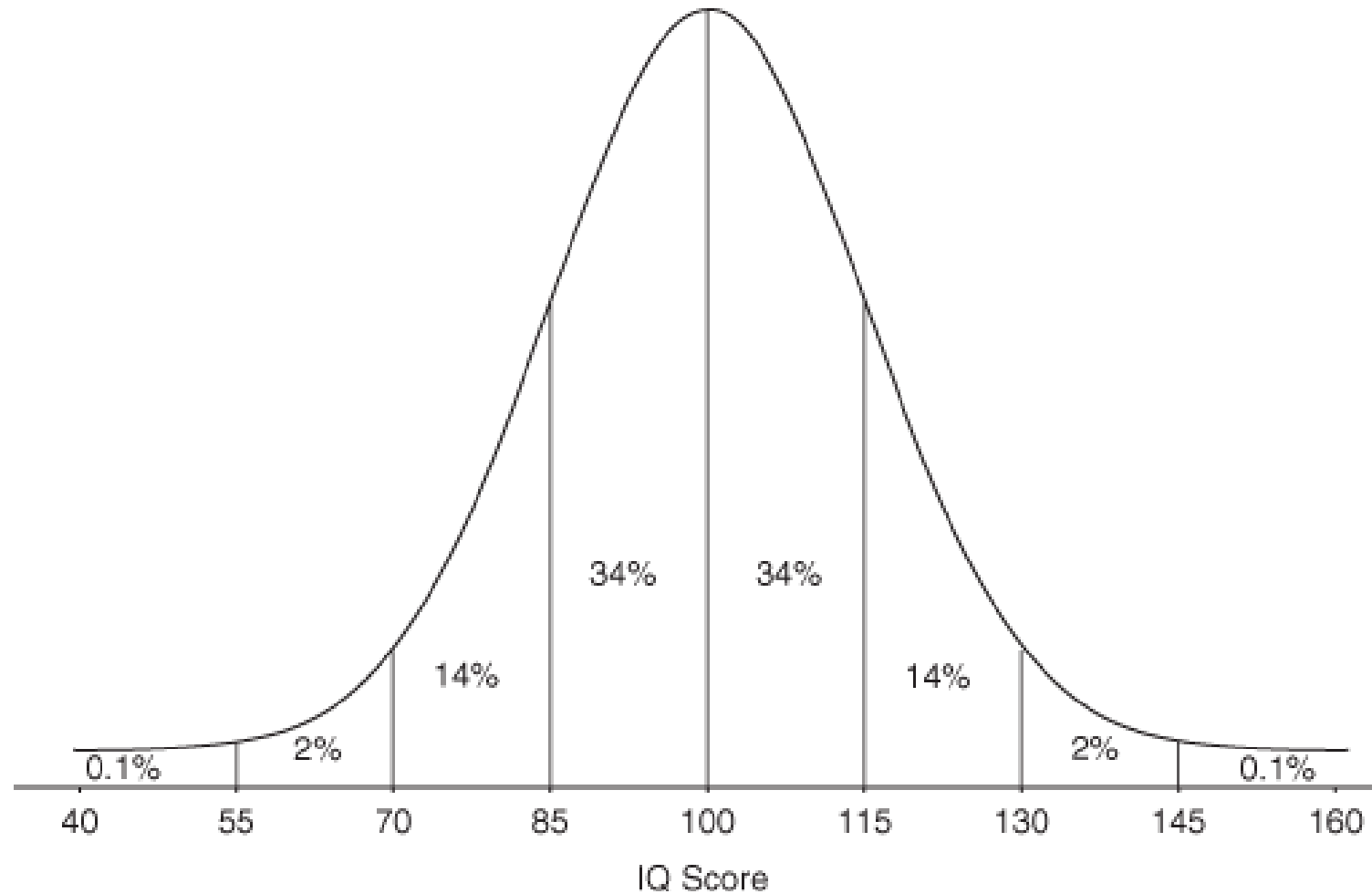
‘g-factor’ or ‘g’

- Is a statistical result
- Used to refer to:
 - general intelligence*
 - general cognitive ability*
 - general mental ability*
- Some posit g-factor is more biological / genetic
e.g. *Haier (2022)*

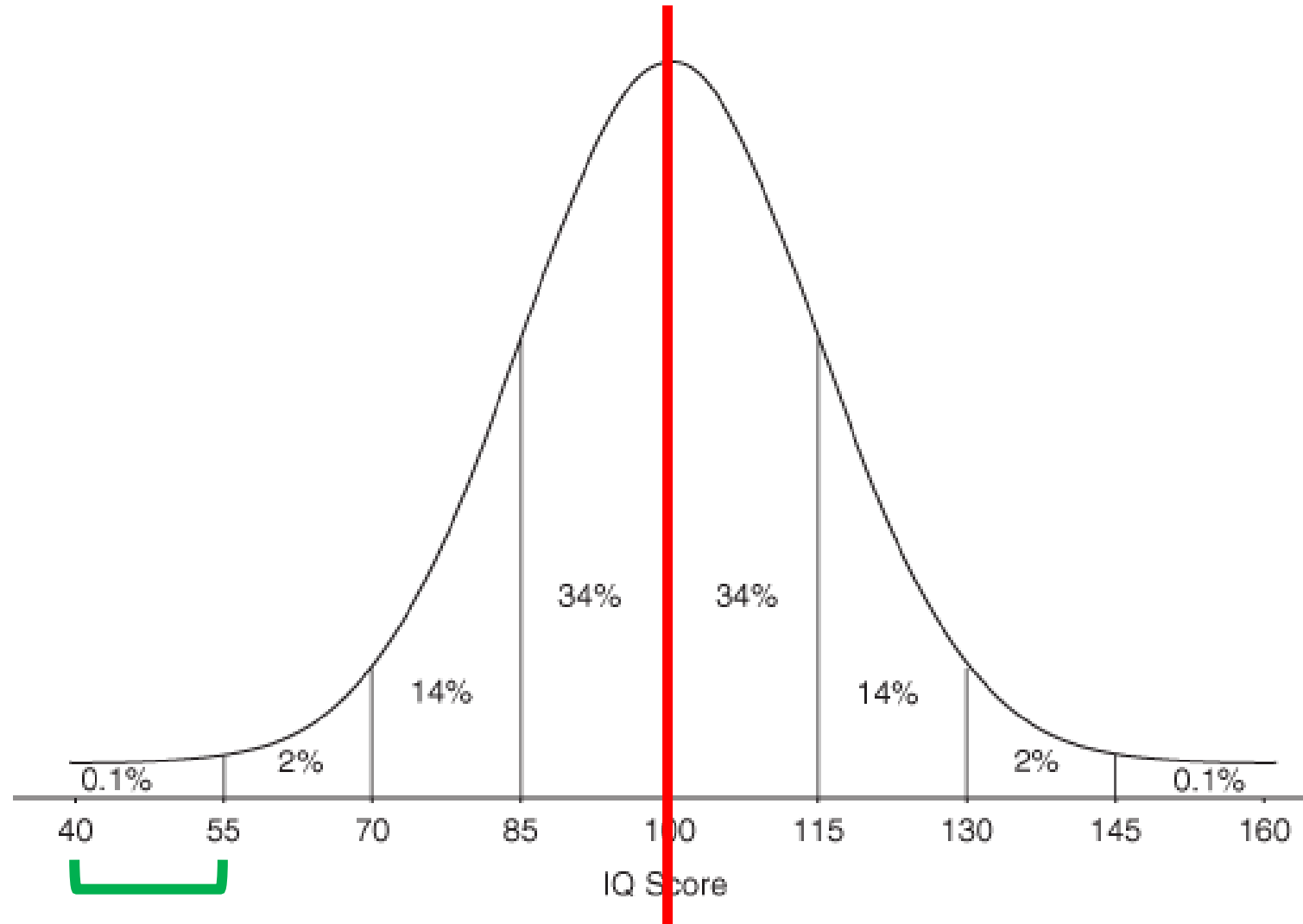
‘Intelligence Quotient’ or ‘IQ’

- Single score representing **range of mental abilities**
- Can be influenced by social & cultural factors, for example
- Large sample of IQ scores are ***normally distributed***

Normal distribution of IQ Scores – ‘the bell curve’

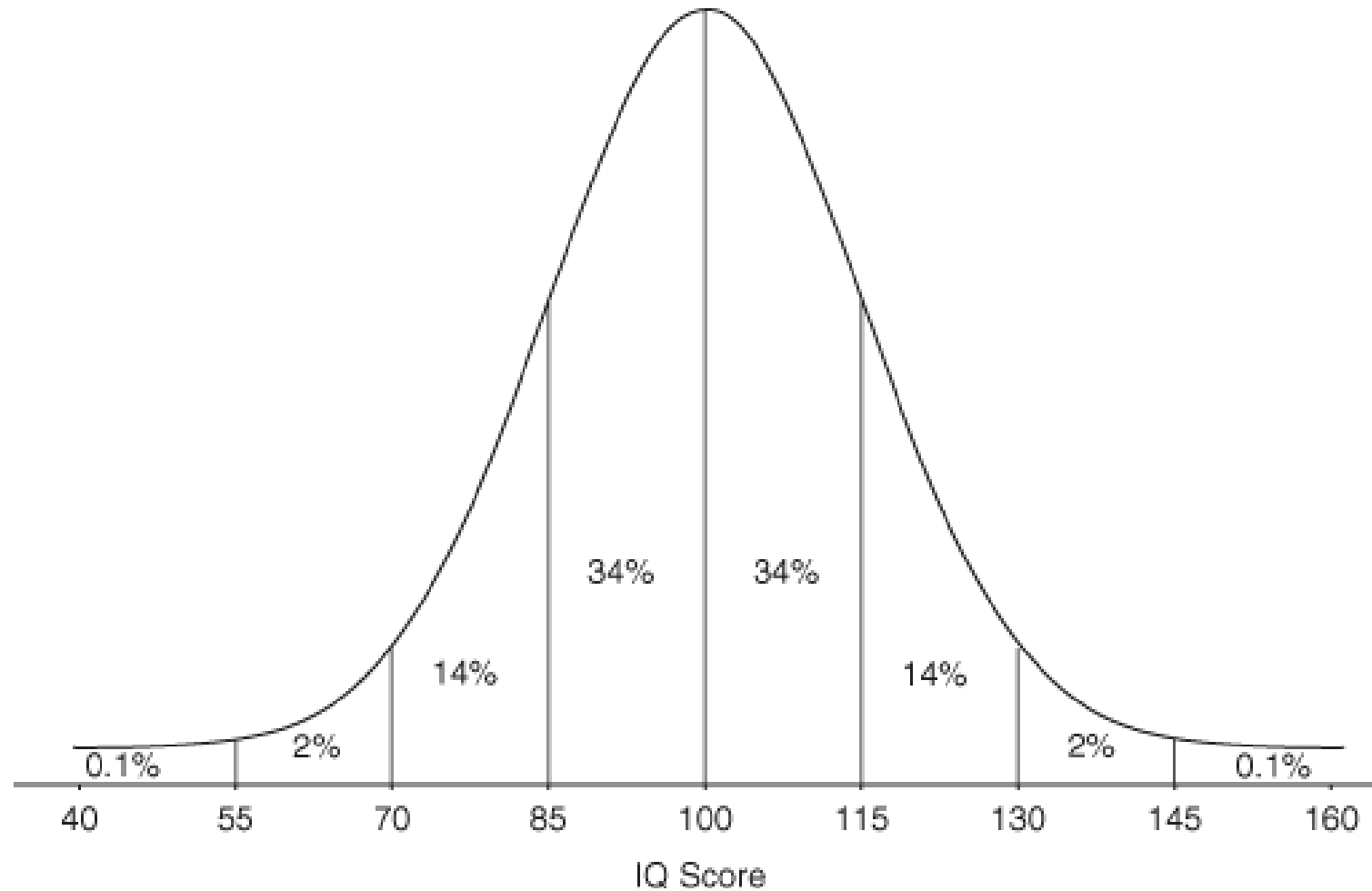


Normal distribution of IQ Scores – ‘the bell curve’



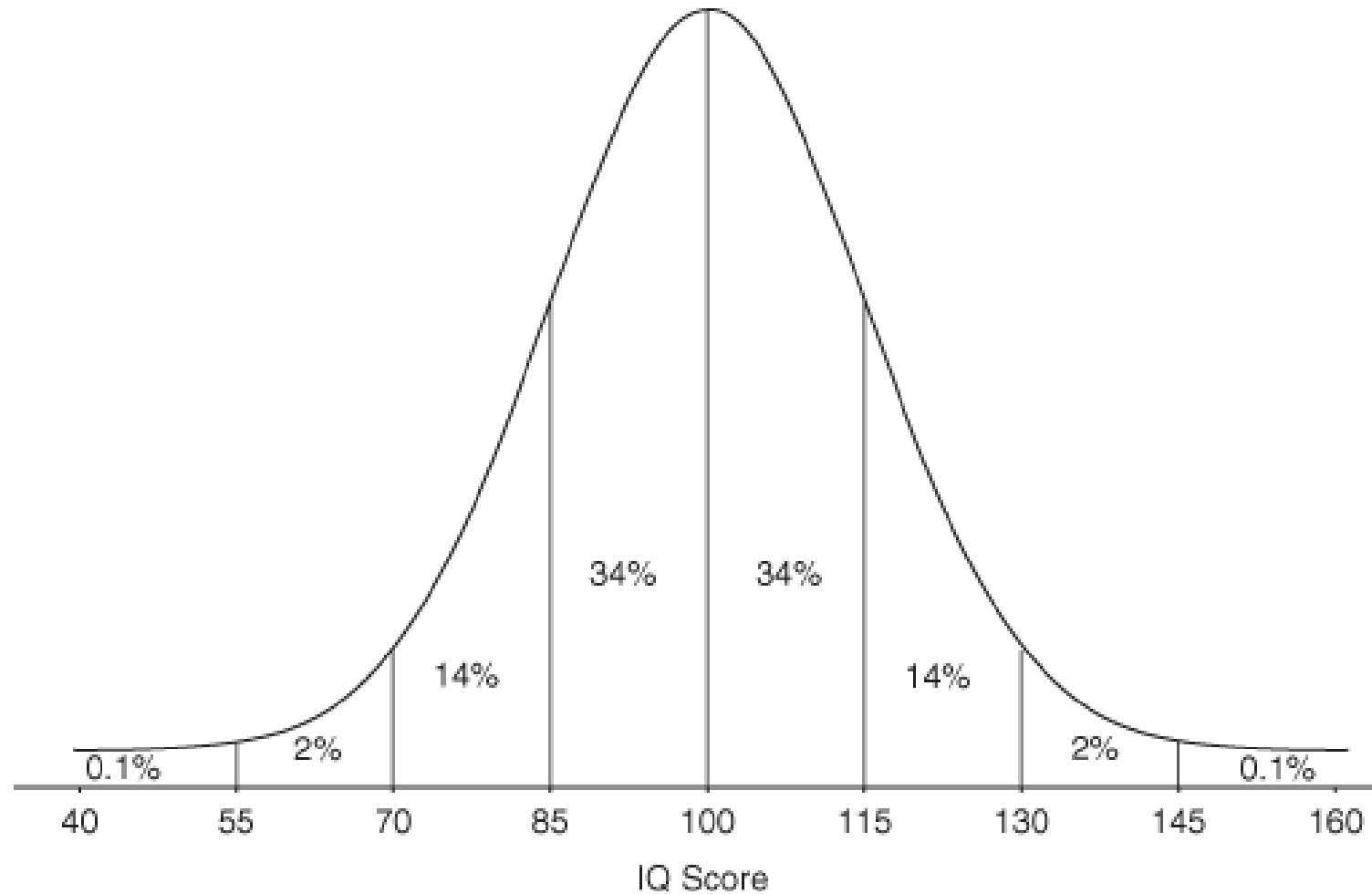
Standard Deviation = 15 Mean = 100

Normal distribution of IQ Scores – ‘the bell curve’



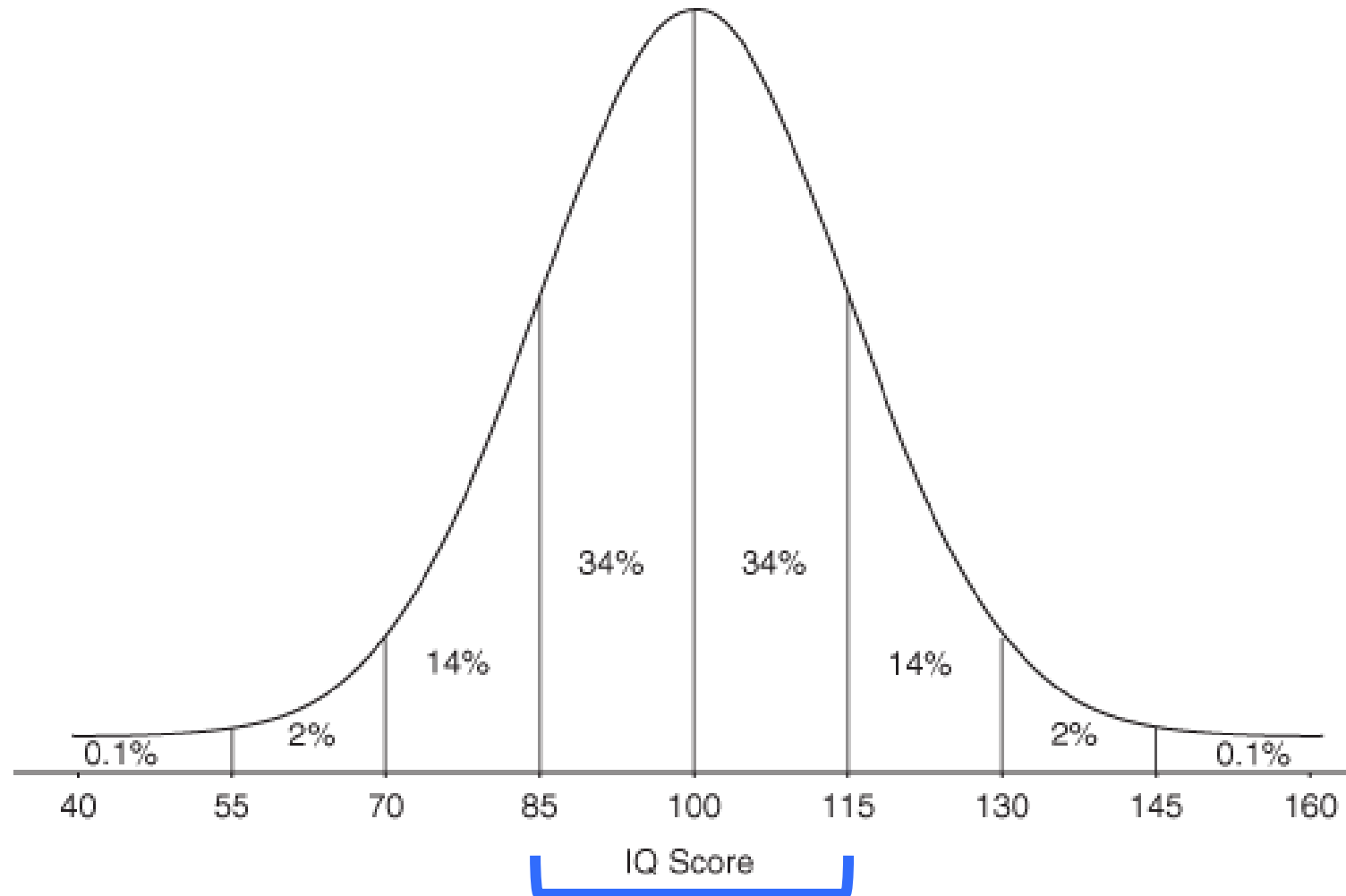
50% of ppl. score >100

Normal distribution of IQ Scores – ‘the bell curve’



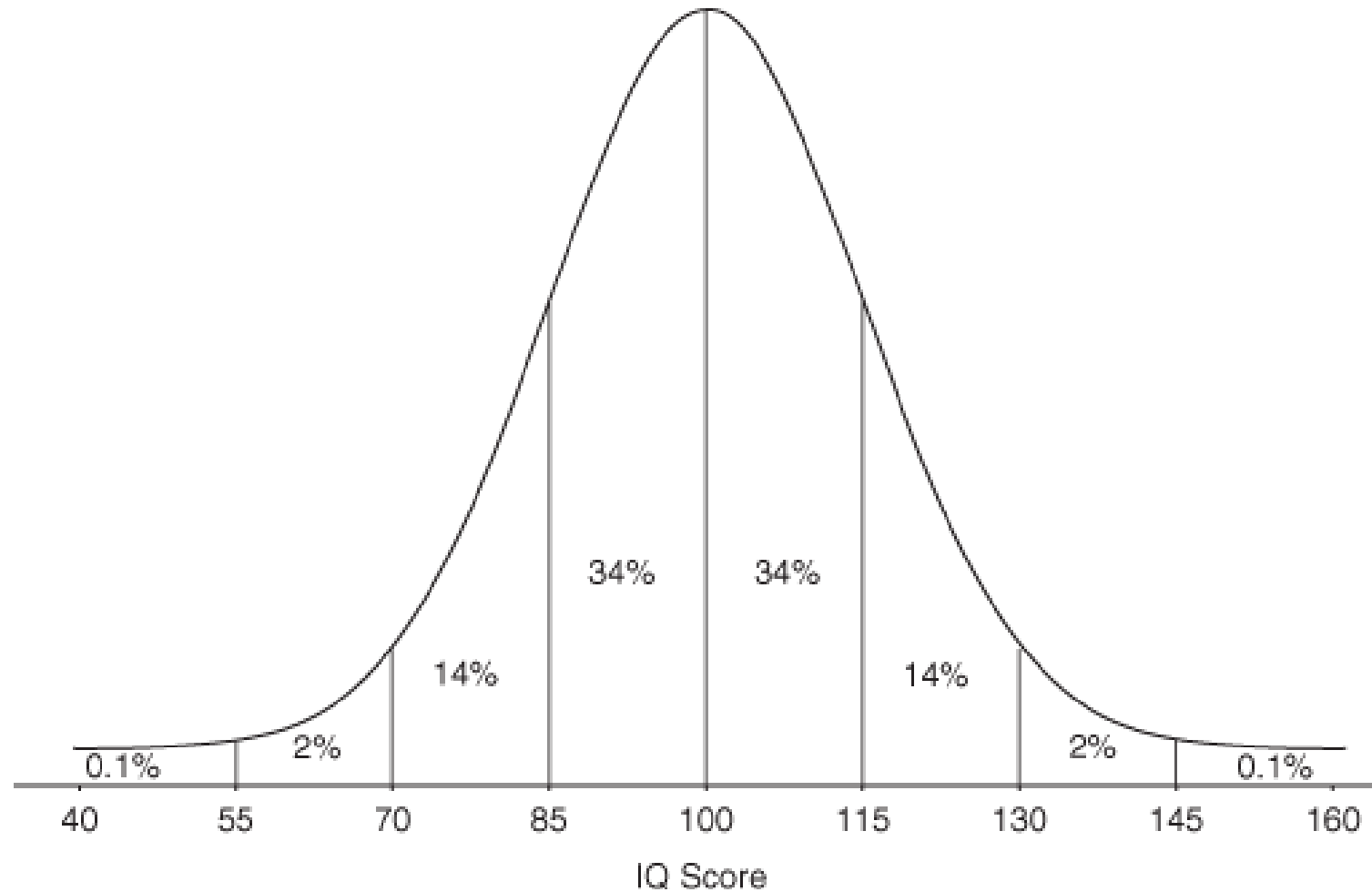
50% of ppl. score < 100

Normal distribution of IQ Scores – ‘the bell curve’



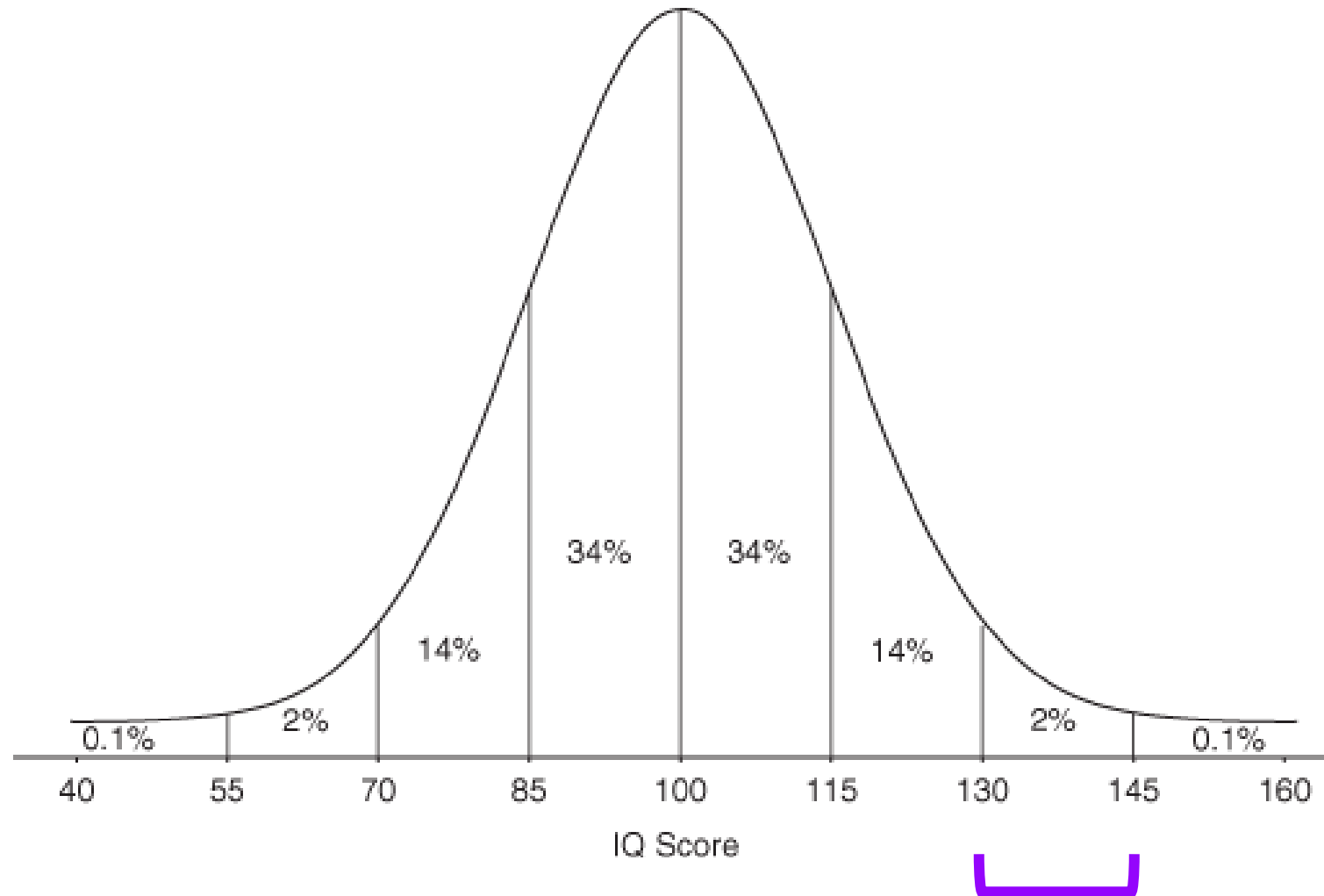
Average IQ between 85-115
68% people

Normal distribution of IQ Scores – ‘the bell curve’



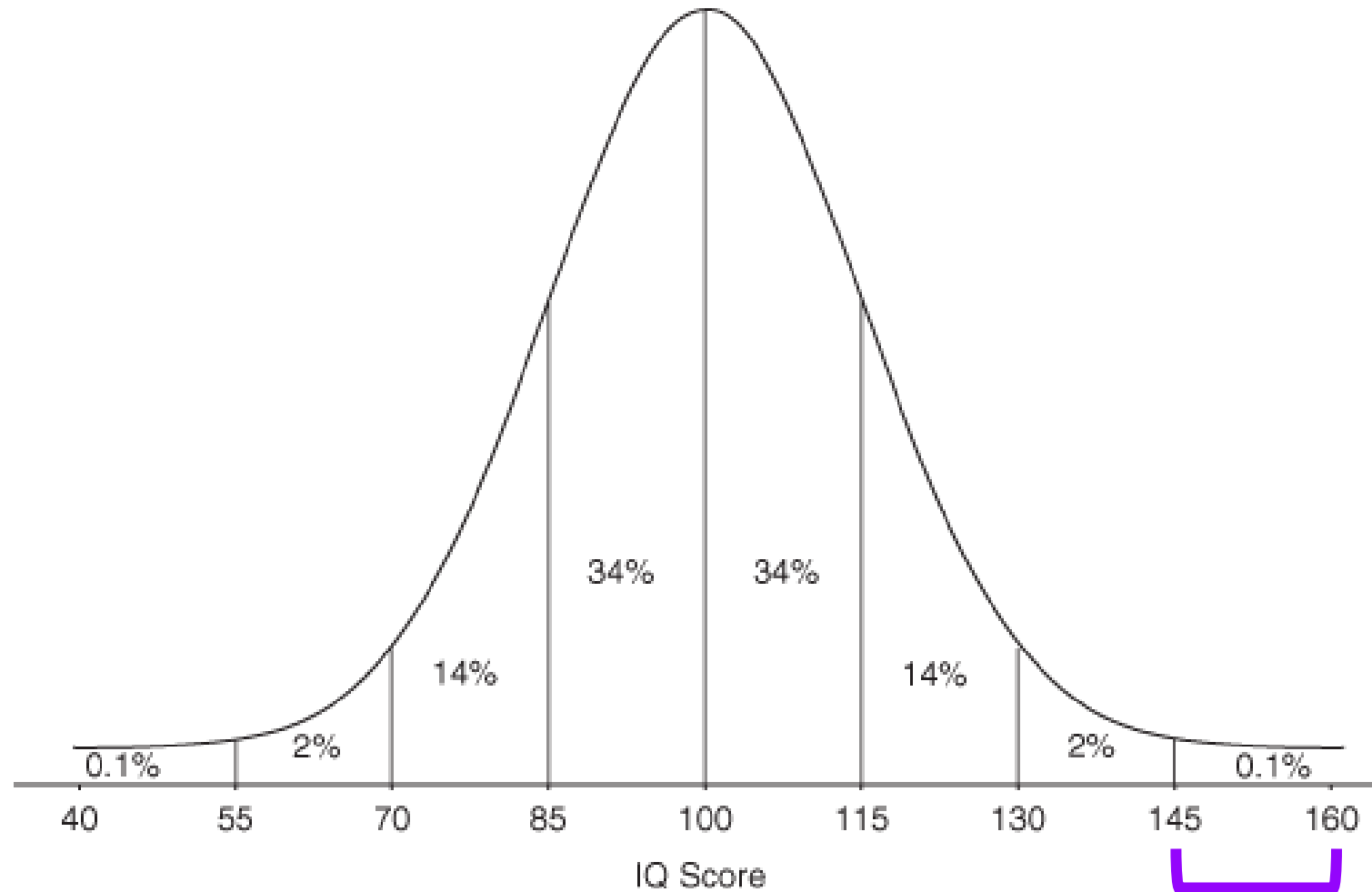
16% of ppl. score <85

Normal distribution of IQ Scores – ‘the bell curve’



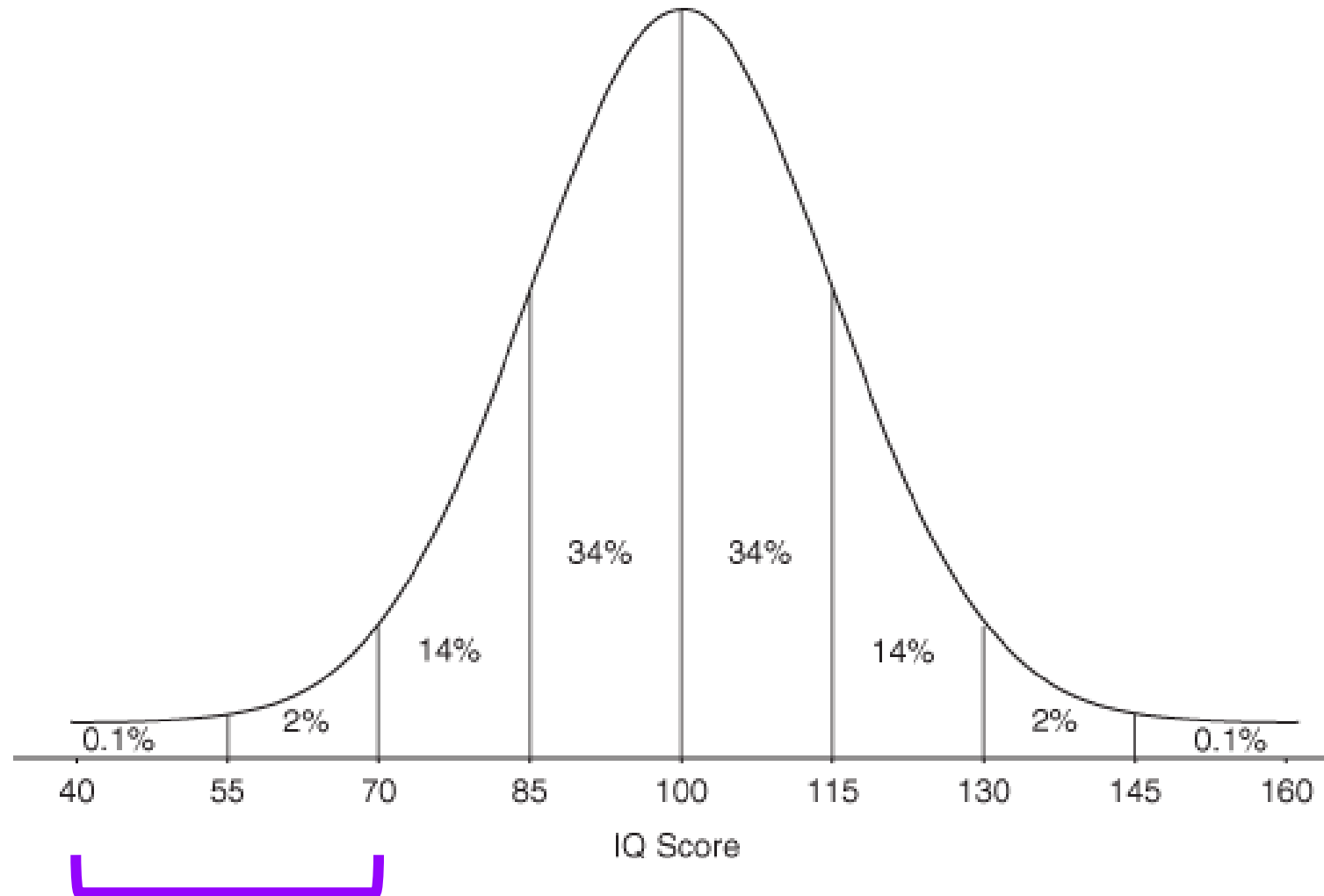
2% of ppl. score between 130-145

Normal distribution of IQ Scores – ‘the bell curve’



> 1% of ppl. score above 145
Classed as 'gifted'

Normal distribution of IQ Scores – ‘the bell curve’

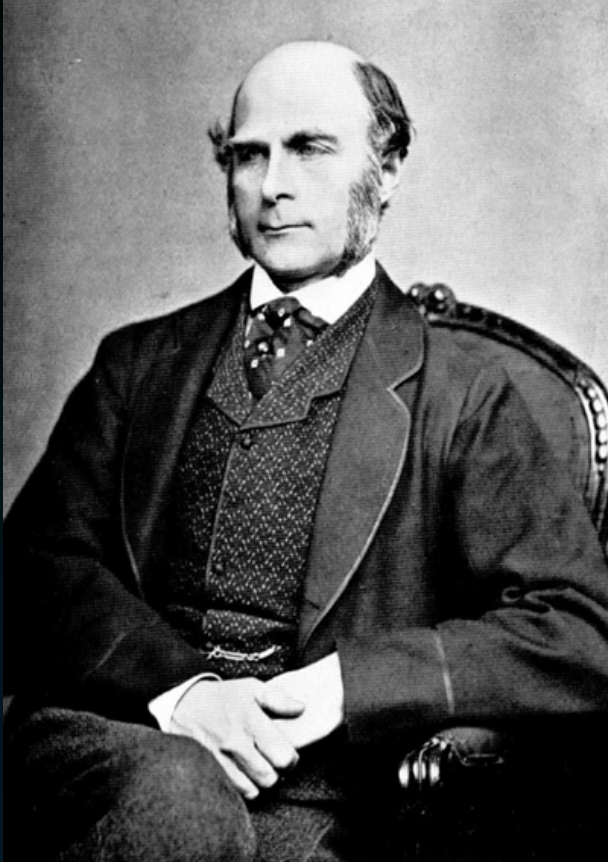


About 2.5% of ppl. score below 70

Early theories of Intelligence

- Understanding the development of IQ tests will help you to gain a better understanding of what these tests are (and what they are NOT).

The Birth of the Psychology of Intelligence

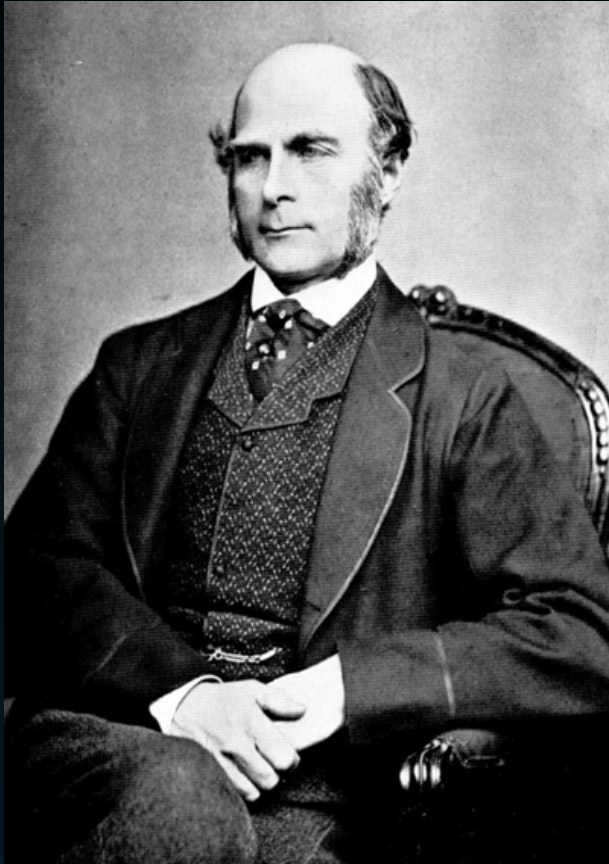


Sir Francis Galton
(1822 – 1911)

- Founder of Differential Psychology
- Founder of Experimental Psychology (together with Wilhelm Wundt, 1832-1920)
- Galton's interest: showing that individuals differ in intelligence.
- Intelligence = dealing with information gained through the senses ('idiotic' people are unable to distinguish colours, heat and cold, etc.).
- Forefather of intelligence test
(response to sensory information, ***reaction times***)

The Birth of the Psychology of Intelligence

- Galton wanted to *measure differences in mental ability* – thereby founded psychometrics

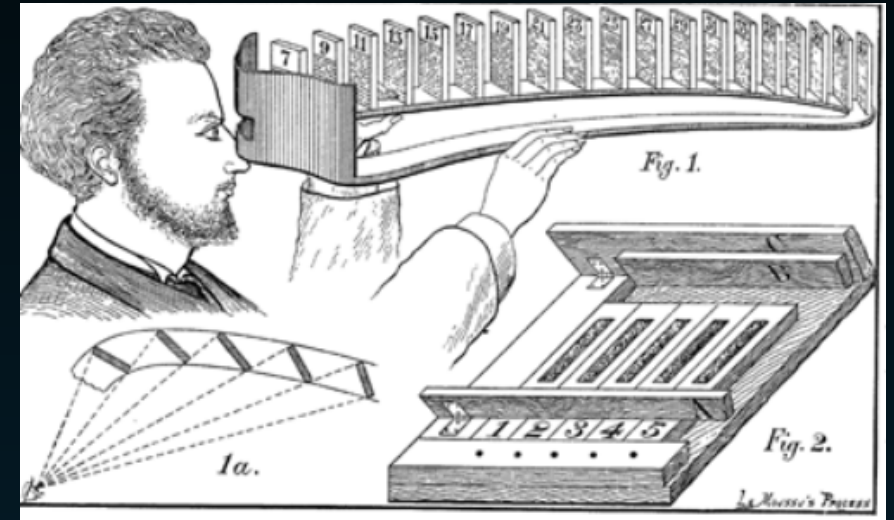


Sir Francis Galton
(1822 – 1911)



"Anthropometric Laboratory", London 1885

9000 sets of data for different ppl. to obtain representative sample
Galton figured his statistical analyses would only be reliable with large sample





James McKeen Cattell
(1860-1944)

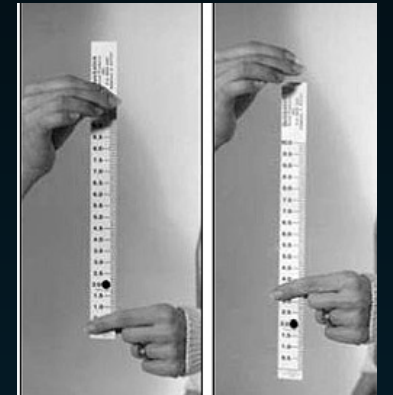
- Taking Galton's view on the importance of the 'senses' for intelligence
- Began to measure "simple mental processes"
- He published a provisional list of 10 'mental tests' designed to measure individual differences



Sensation

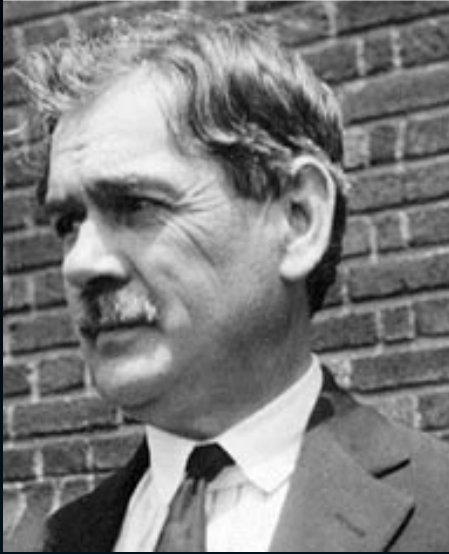


Weight discrimination



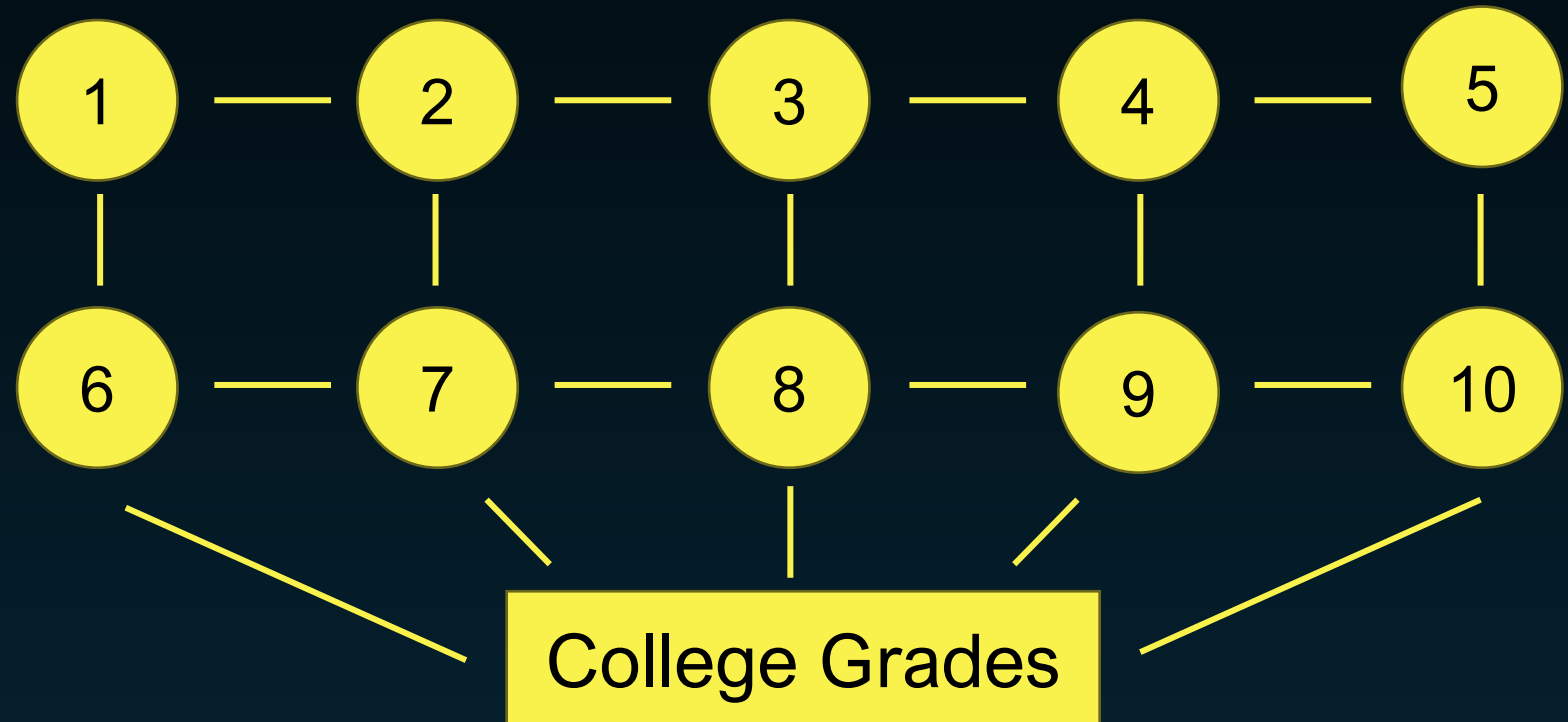
Reaction
Times

The Fall of Cattell's 'mental tests'

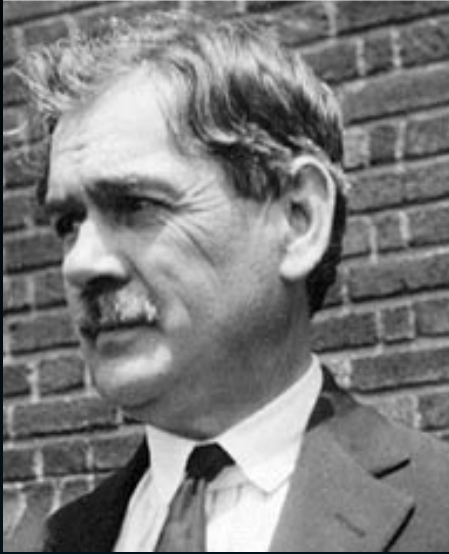


Clark Wissler
(1870-1947)

- Wissler (1901) using new correlation stats perfected by Karl Pearson (from Galton's ideas) compared Cattell's 'mental test' scores in UG's with their college grades.

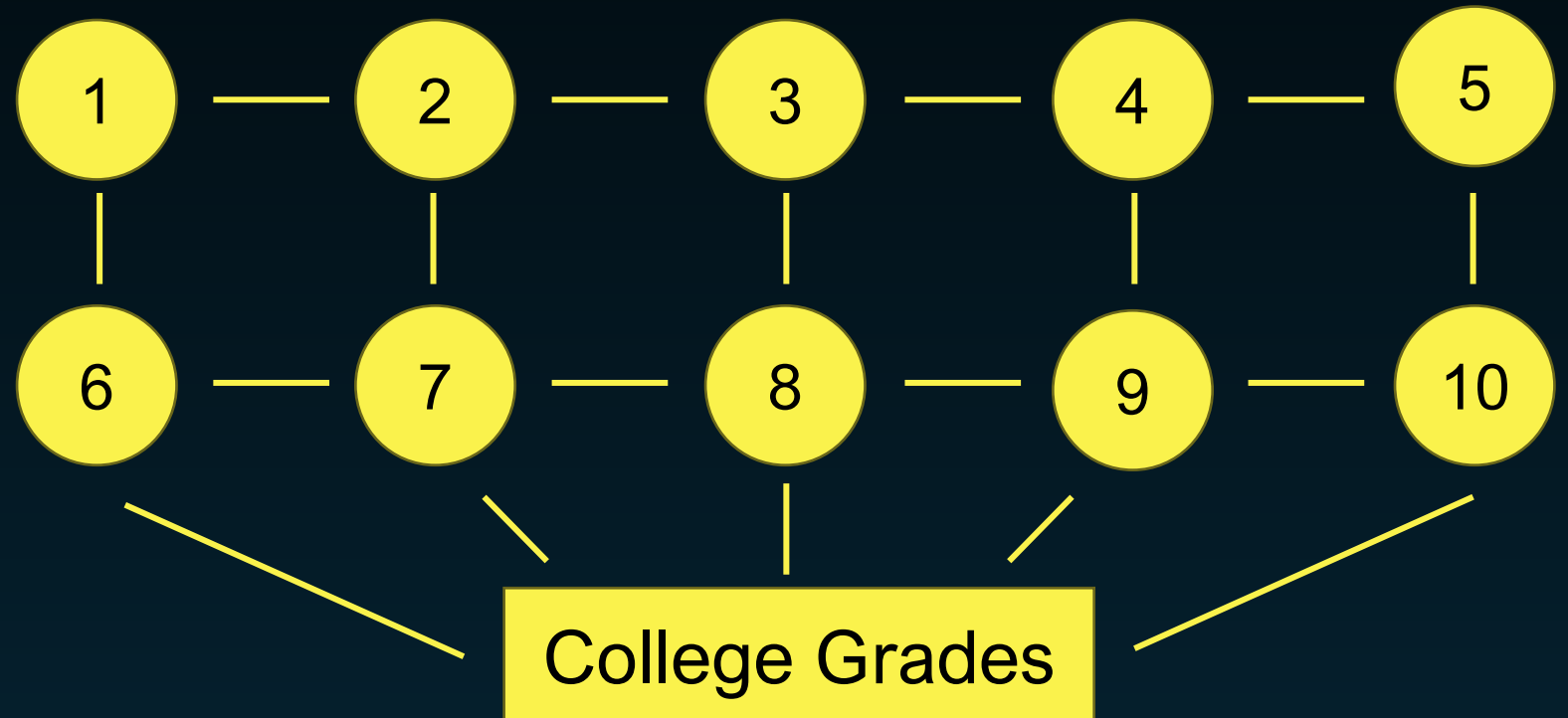


The Fall of Cattell's 'mental tests'



Clark Wissler
(1870-1947)

- No correlation between any of Cattell's mental tests
- Crucially, none of Cattell's tests correlated with college grades (though grades on one course correlated well with grades on others)



Take home from Galton & Cattell

When trying to devise a test of intelligence, one must have some independent criterion of intelligence with which the test can agree....



Alfred Binet
(1857 – 1911)

- French government commissioned Binet to design an intelligence test for children to identify students requiring alternative educational provision
- Pragmatic approach, relying more on practicality, common sense, coping with the day-to-day world





Alfred Binet
(1857 – 1911)

- 1905 Binet created the first intelligence test (for children)
- The **Binet-Simon scale** included 30 short everyday-related tasks:
 - Following a light match with your eyes
 - Naming parts of the body
 - Counting coins
 - Naming objects in a picture
 - Recalling a number of digits after being shown a long list
 - Word definitions
 - Filling in missing words in a sentence

Binet-Simon scale used children's age as an independent criterion of intellectual competence



4-year-olds



6-year-olds



8-year-olds

intellectual competence

Binet-Simon scale used children's age as an independent criterion of intellectual competence

Test construction involved continuing experimentation and refinement



Example Task: Copy a square on a piece of paper



4-year-olds



6-year-olds



8-year-olds

Binet-Simon scale used children's age as an independent criterion of intellectual competence

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Binet-Simon scale used children's age as an independent criterion of intellectual competence

Test construction involved continuing experimentation and refinement



Example Task: Copy a square on a piece of paper



4-year-olds



6-year-olds



8-year-olds



This appears as an item on 4-year-old task (majority of 4-year-olds and all older kids can do it)

Binet-Simon scale used children's age as an independent criterion of intellectual competence

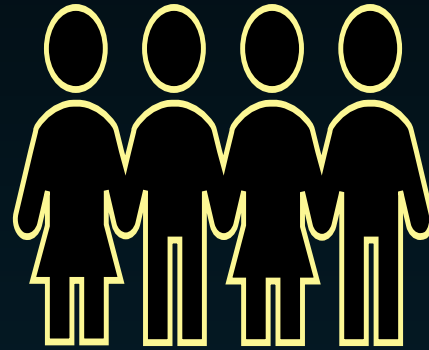
Test construction involved continuing experimentation and refinement



Example Task: Copy a diamond on a piece of paper



4-year-olds



6-year-olds



8-year-olds

Binet-Simon scale used children's age as an independent criterion of intellectual competence

Test construction involved continuing experimentation and refinement



Example Task: Copy a diamond on a piece of paper



4-year-olds



6-year-olds



8-year-olds



Binet-Simon scale used children's age as an independent criterion of intellectual competence

Test construction involved continuing experimentation and refinement



Example Task: Copy a diamond on a piece of paper



4-year-olds



6-year-olds



8-year-olds



This appears as an item on 6-year-old task (not on the 4 or 8-year-olds)

Binet-Simon scale used children's age as an independent criterion of intellectual competence

Final set of tests (Binet & Simon 1911) from 3 – 10 years of age



3

4

5

6

7

8

9

10



Dennis passed the 7-year-old but failed the 8 year-old test = 'mental age' of 7

Jenny passed the 9-year-old but failed the 10-year-old test = 'mental age' of 9



Binet-Simon scale used children's age as an independent criterion of intellectual competence

Binet used a *relative* or *normative* measure of intelligence – a mental age of 6 was the average score obtained by lots of 6-year-olds

A 5-year-old with a mental age of 6 was advanced, but a 10-year-old with a mental age of 6 was behind.

Dennis passed the 7-year-old but failed the 8 year-old test = 'mental age' of 7

Jenny passed the 9-year-old but failed the 10-year-old test = 'mental age' of 9

Binet-Simon scale used children's age as an independent criterion of intellectual competence

The Binet Scales formed the basis for modern IQ tests, just as mental age formed the basis for IQ scores

The use of **age** in psychological testing is one of Binet's lasting contributions

Dennis passed the 7-year-old but failed the 8 year-old test = 'mental age' of 7

Jenny passed the 9-year-old but failed the 10-year-old test = 'mental age' of 9

Taking the concept of mental age one step further



William Stern
(1871 – 1938)

- Coined the term intelligence quotient (IQ)
- To see if child is developing normally – examine the ratio of child's *mental* age to *physical* age

IQ Score

$$(\text{Mental age} \div \text{Physical age}) \times 100$$

Taking the concept of mental age one step further



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IQ Score

$$(\text{Mental age} \div \text{Physical age}) \times 100$$

Averaged across all subtests

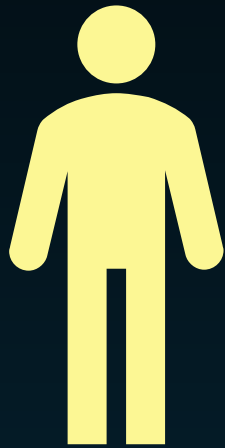
Avoids dealing with fractions

Taking the concept of mental age one step further



William Stern
(1871 – 1938)

- Coined the term intelligence quotient (IQ)
- To see if child is developing normally – examine the ratio of child's *mental* age to *physical* age



John is 10-years-old

John's test score same as
average 8-year-old



$$(8 \div 10) \times 100 = 80$$

John has an IQ of 80

However...concept of mental age is hard to assess after
16 years of age



Alfredo
21-years-old



Sherri
23-years-old



Deepika
32-years-old

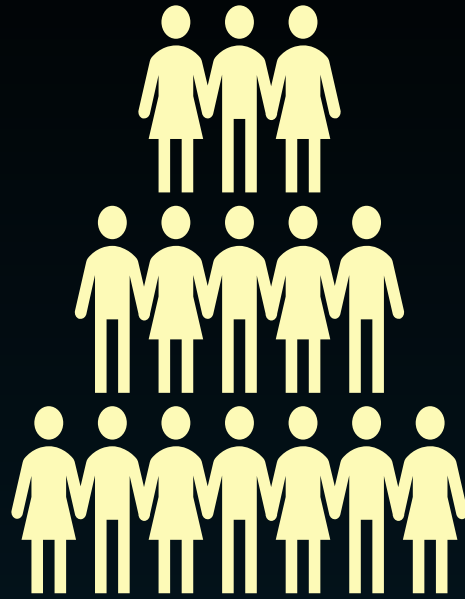


Klein
40-years-old

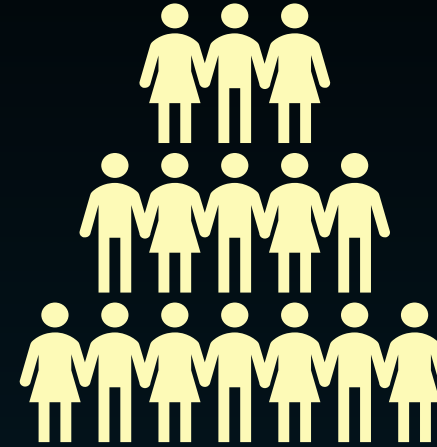
IQ scores are only meaningful relative to people of same age



Sherri
23-years-old
IQ of 120



Scores above
average relative
to her peers
(age matched)



Scores above
average relative
to her peers
(age matched)



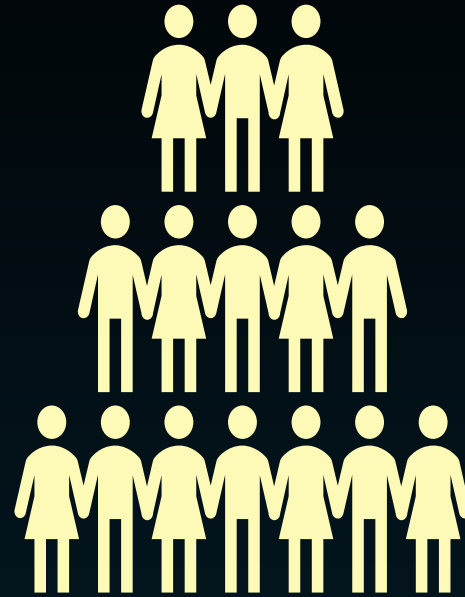
Kesha
12-years-old
IQ of 120



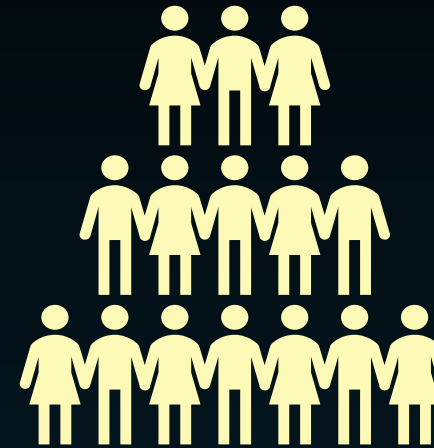
IQ scores are only meaningful relative to people of same age



Sherri
23-years-old
IQ of 120



Scores above
average relative
to her peers



Scores above
average relative
to her peers



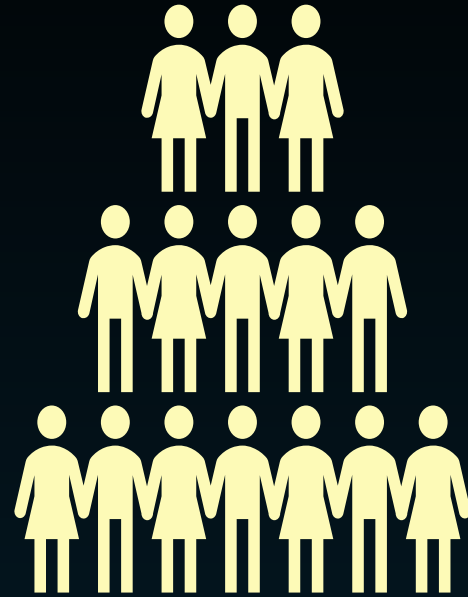
Kesha
12-years-old
IQ of 120

Not accurate to say Sherri and Kesha are equally intelligent

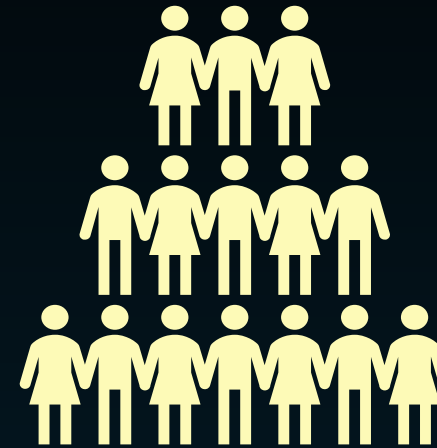
IQ scores are only meaningful relative to people of same age



Sherri
23-years-old
IQ of 120



Scores above
average relative
to her peers



Scores above
average relative
to her peers



Kesha
12-years-old
IQ of 120

Not accurate to say Sherri and Kesha are equally intelligent

Take home from IQ Scores

- IQ scores are not *absolute* measures, like centimetres of height or kilometres of distance
- IQ is an *ordinal* measure where we are simply rank-ordering people
- There is no zero for IQ, as there is for height or distance
- Someone with an IQ of 120 is not twice as intelligent as another with an IQ of 60
- To say that someone has an IQ of 130 is to say their score lies in the top 2.5% of a representative sample of people of the same age

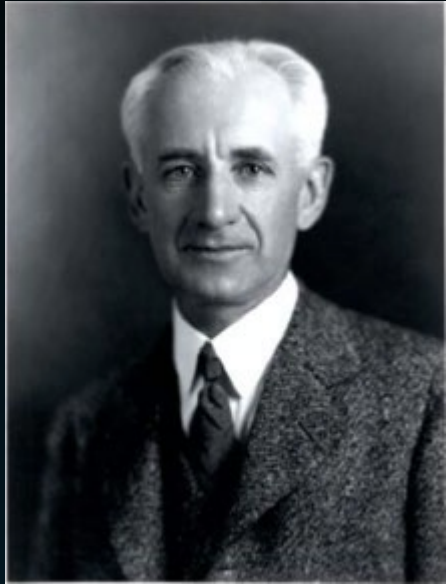
Stanford-Binet Test



Lewis Terman
1877-1956

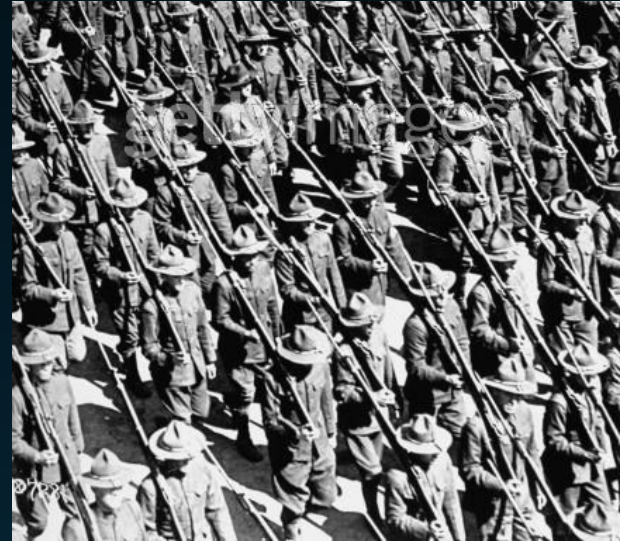
- 1916: Lewis Terman produced a Stanford revision of the Binet-Simon scale:
- the Stanford-Binet test (now on 5th edition) still remains one of the standard tests of intelligence to this day
- Virtually a new test for Binet's with forty new items added with far more children tested in each age group
- However, these tests with trained examiners were clunky to administer and certain answers open to interpretation

Stanford-Binet and Army tests...growth of an IQ industry

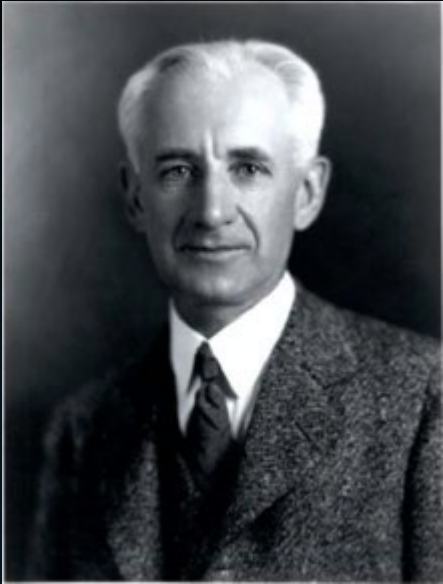


Robert Yerkes
(1876 – 1956)

- Yerkes with the help of Terman and Goddard designed a new IQ test for use by US army
- 2 tests - alpha and beta
- **Questions were timed and multiple-choice style**
- Beta test designed for illiterate groups
- Simple tasks as quickly as possible

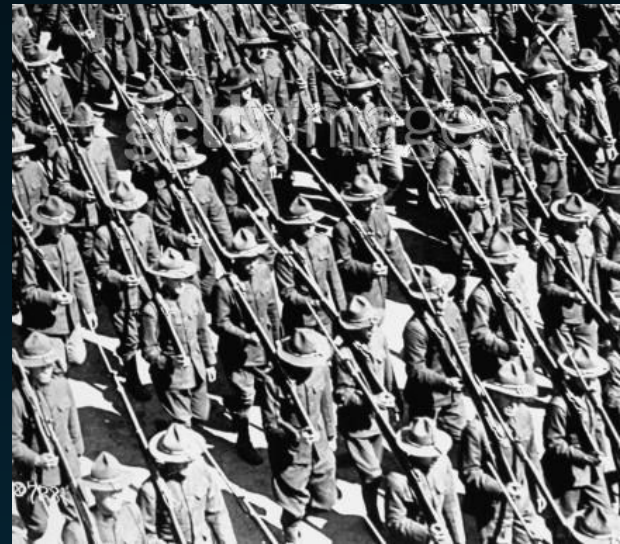


Stanford-Binet and Army tests...growth of an IQ industry



Robert Yerkes
(1876 – 1956)

- The army tests transformed public opinion to mental tests
- Yerkes published a new Intelligence Test in 1919
- Tests were used everywhere (schools, universities, business firms, clinical settings)
- **1.75M were tested – huge sample**



Final thoughts on logic of generating an operational definition

Example: Want to study aggressiveness in children



Example
adapted from
Shacter

Operationally defined as 'shoving others'

Final thoughts on logic of generating an operational definition

Follow children around for one week and count number of shoves?

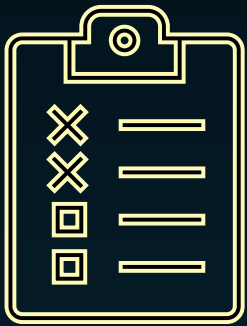


This type of measurement would be impractical to implement

Final thoughts on logic of generating an operational definition

Better solution might be to develop a set of questions or tasks highly correlated with shoving

Questions



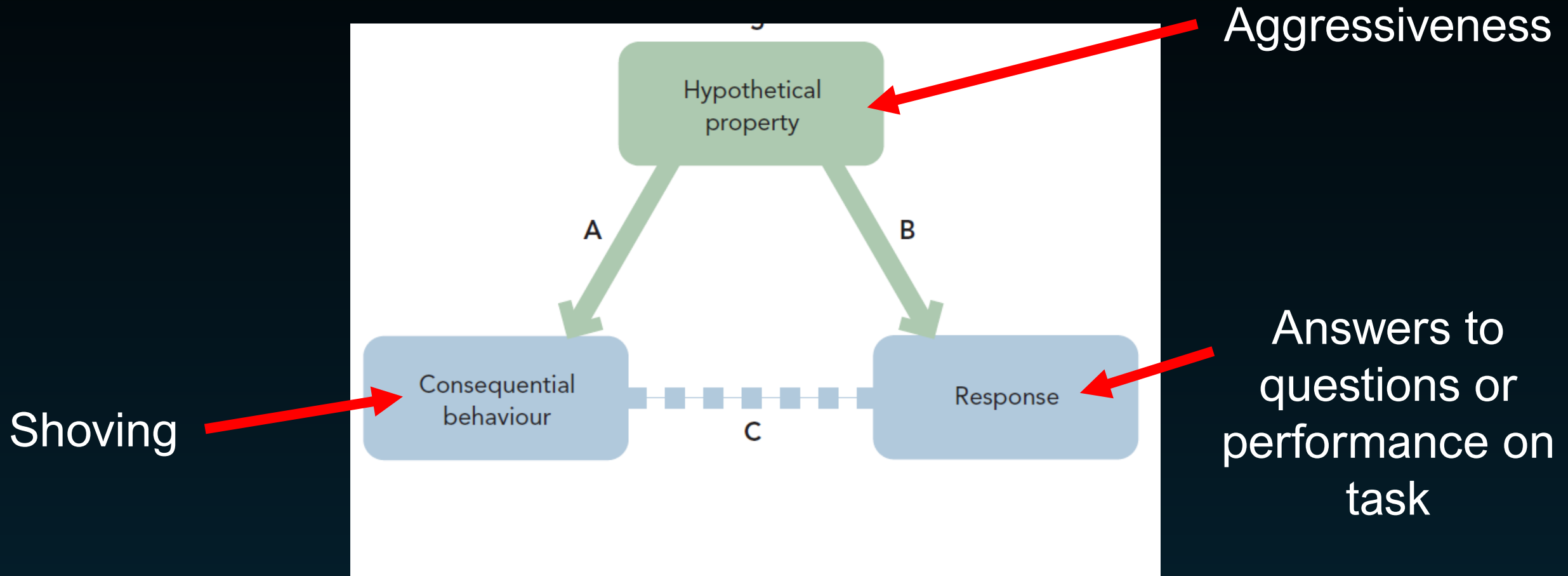
Do you think most people need a good punch in the face from time to time?

Tasks

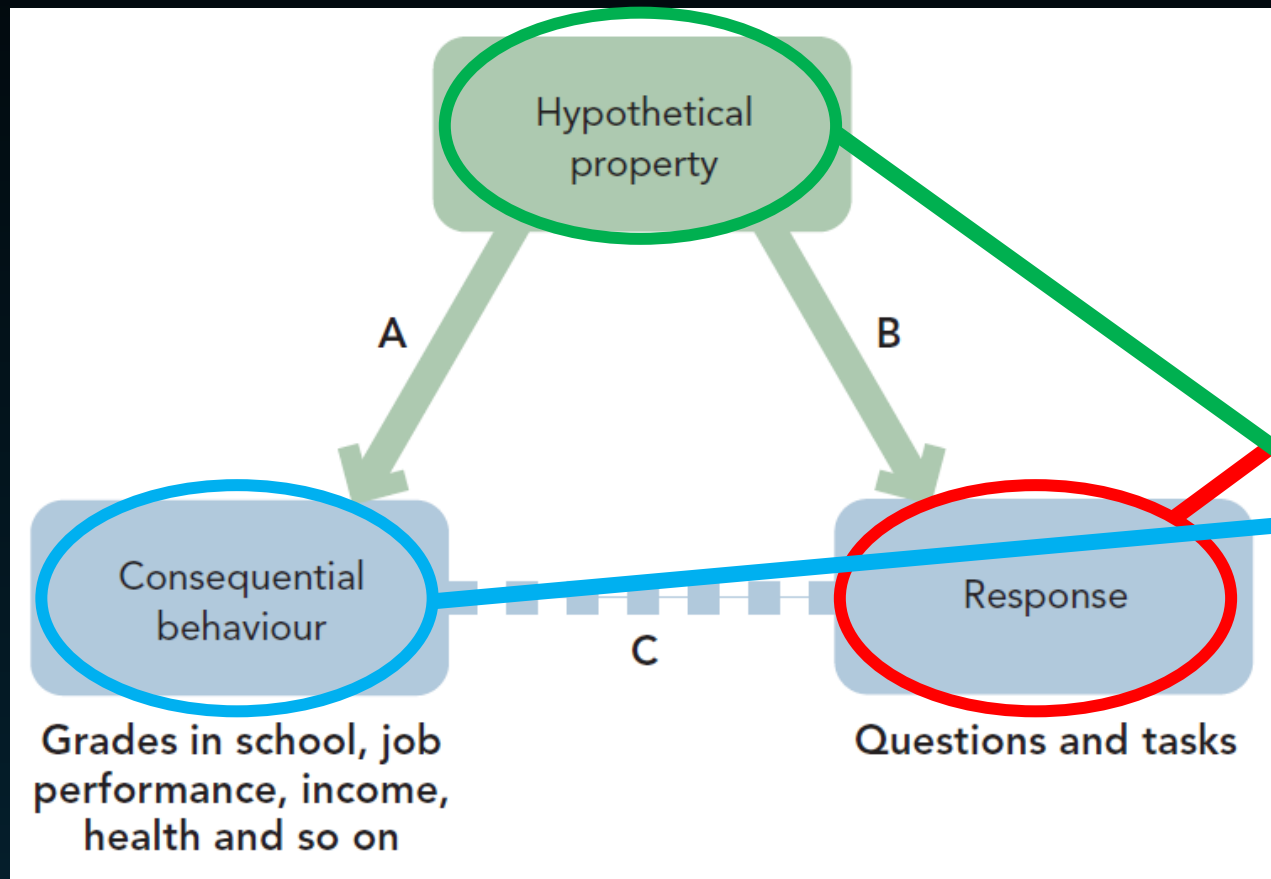


How quickly can you decapitate this doll?

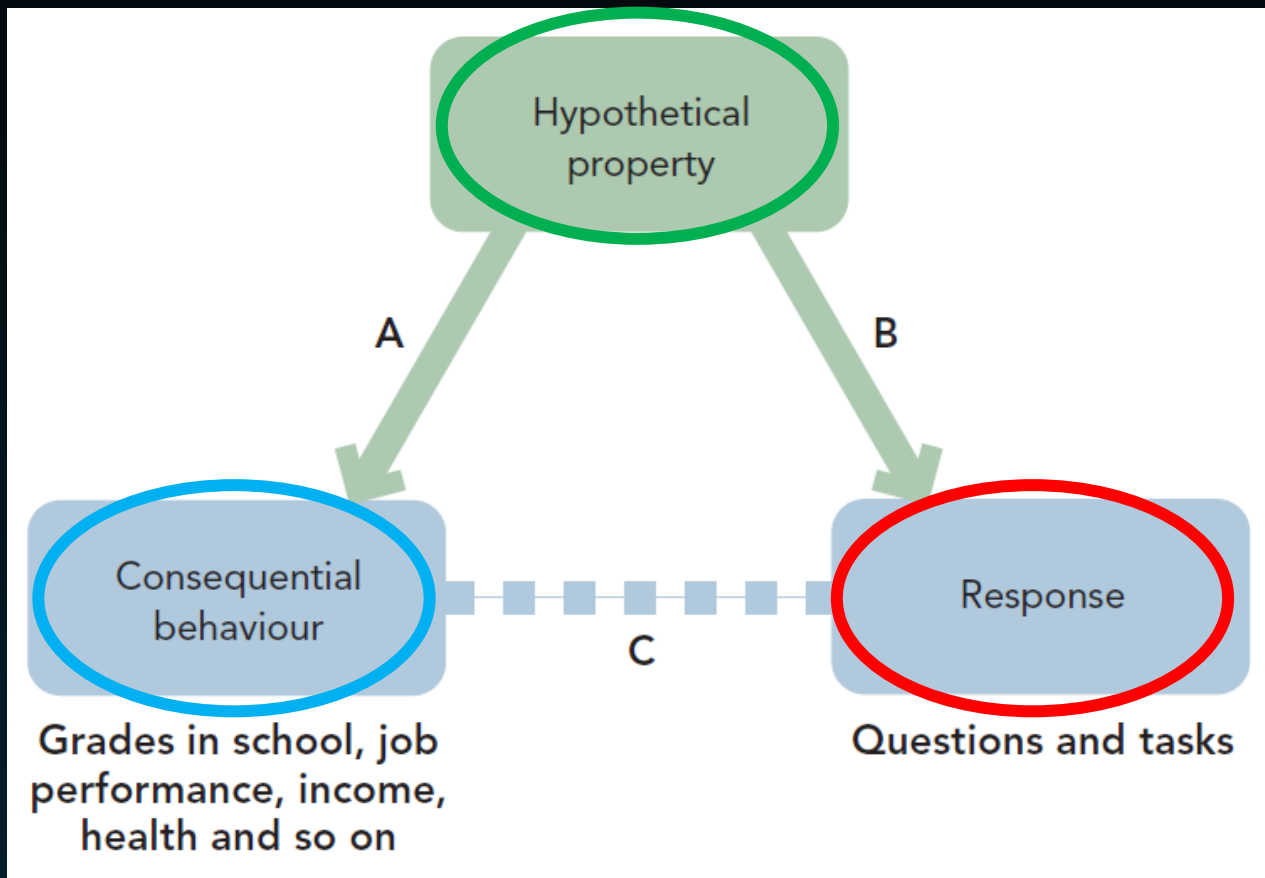
Final thoughts on logic of generating an operational definition



Final thoughts on logic of intelligence testing – generating operational definition



Final thoughts on logic of intelligence testing – generating operational definition



- Measuring consequential behaviour is usually not very efficient
- Using a test has several advantages:
 - Easy to do
 - Less time consuming
 - Larger samples
 - Standardised

Summary

- Extraordinary but narrow mental ability, e.g. savant syndrome – not usually considered evidence of high intelligence
- Specific mental abilities are not independent – if you do well on one, you tend to well on others
- All these test scores have something in common – the general factor of intelligence or the ‘g-factor’
- IQ scores tend to represent a range of mental abilities and are normally distributed
- An independent criterion, e.g. college grades, is important for an intelligence test to be compared against
- Age is important to consider in IQ scores, especially in children
- Intelligence tests – should be easy to administer and standardise